Data Security Project - Scenario and Project Description

Data Security Project Student Checklist

This is a Group piece of work but you can work it on on your own if you do not want to be part of a group. I **strongly** advise you to do it as a group project.

This project helps you prepare for the **exam** and should give you a useful resource later.

It will not be marked but I would really like you to work on it alongside your portfolio work and would really welcome the chance to look at it and discuss it with your group.

The portfolio is a collection of individual artefacts but the project report puts them all together and gives you a single source in the future for data security information.

The Project Scenario

Your team are acting as a **data security consultancy team** and your job is analysing and improving the data security of a **university student and staff management information system**.

The system handles multiple types of data, including:

- student records
- grades and feedback
- staff information (personnel information, career tracking MUS {medarbejdere udviklingssamtale}data)
- research-related data (data sets which can be in a data base, text files, images, video etc)
- audit and system data (system and application logs etc)

The university operates a **mixed environment**, including:

- on-premise systems
- cloud-based services
- web-based access for students and staff

Your task is to **analyse risks, propose security controls, and justify design decisions** related to how data is:

- accessed
- stored
- modified
- audited
- protected during use and transmission
  You will need to think about the type of systems, explore solutions and explain the choices you have made

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Project Goals

The project aims to demonstrate that you can:

- apply **core data security concepts** in a realistic scenario
- reason about **confidentiality, integrity, and availability**
- design and justify **access control and protection mechanisms**
- understand the **limitations of technical controls**
- communicate security decisions clearly and professionally

Technology Guidance

This is **NOT a programming-heavy project** but you are encouraged to use tools (Threat Dragon for example), code (example scripts for potential risks - so you might show code that monitors logs to report unusual login activity like multiple failed attempts),

You are expected to:

- demonstrate understanding of tools and techniques
- explain ***why*** particular solutions are appropriate

-You can provide **code snippets or scripts** where relevant as screenshots in the report or as additional items.

Recommended

- **Python** as this is the language you learned in 1st semester and is widely used in data analysis work
  - hashlib (hashing)
  - cryptography (encryption)
  - PyJwt (generating tokens)
  - secrets (generating strong passwords)

-**Bash scripts** - good for combining and running *nix commands such as monitoring logs or checking file integrity.

-**Other languages if you want** I reccommend Go, Rust, C, or Zig for these types of system focused tasks and learning C will never be a waste of time.

Important Constraints

- Avoid exotic or experimental technologies - use tools that you might find in industry so Elastic Search for log monitoring for example
- Use **well-documented, widely adopted libraries**
- Focus on **correct usage and explanation**, not performance or optimisation
- Full applications are **not required** but you will be expected to do research and be willing to have a go at script / code writing to help show how you would handle risks you have identified

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Project Deliverables

1. Written Report (15–20pages)

- Approximately **2400 characters (including whitespace) per page**
- Language: **Danish or English** (but remember exam is in Danish)
- Format: PDF

The report must be written in a clear style suitable for a technical but non-specialist audience.

2. Technical Artefacts (Appendix or Repository)

Examples may include:

- small scripts / code examples
- configuration examples
- diagrams
- tables
- screenshots of tools or outputs
- logs or audit examples

Artefacts must be:

- clearly referenced in the report and things like code can be done as screenshots
- explained (what they demonstrate and why they matter)

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Suggested Report Structure

This will ensure you get a clear and coherent report structure that is easy to get information from.

1. Introduction

- Description of the system and scenario
- Scope and assumptions
- Overview of security concerns

2. Threat Modelling & Risk Analysis

- Identification of data assets
- Threat actors and threat scenarios
- CIA impact analysis
- Initial risk assessment

3. Authentication & Identity Management

- User and role definitions
- Authentication approaches
- Risk discussion (credential compromise, misuse)
- Justification of choices

4. Authorisation & Role-Based Access Control

- Role-based access model
- Least privilege considerations
- Risk assessment of role misuse
- Reflection on trade-offs

5. Data Storage, Integrity & Cryptography

- On-prem vs cloud storage considerations
- Hashing vs encryption (when and why)
- Integrity protection
- Basic discussion of SBOM or dependency risk

6. Auditing, Logging & Accountability

- Audit requirements
- Layered audit trail design
- Sensitivity and protection of audit data
- How incidents could be investigated

7. Encryption at Rest & in Transit

- Encryption strategy for stored data
- Whole disk encryption considerations
- TLS and data in motion
- Limitations of encryption

8. Data in Use & Exposure Risks

- Identification of data exposure points
- Token-based access (conceptual)
- Risks during active use
- Non-technical mitigations

9. Governance, Classification & Data Lifecycle

- Data classification
- Retention and deletion policies
- Ethical considerations
- Shared responsibility (especially cloud)

10. Conclusion & Reflection

- Summary of key risks and mitigations
- Remaining uncertainties
- Reflection on limitations and future improvements

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Expectations for the Exam

You should be able to:

- explain the project as a whole
- justify key security decisions
- reflect on trade-offs and limitations
- demonstrate understanding beyond their own section

You may be asked:

- “Why did you choose this approach?”
- “What risks remain?”
- “What would you do differently with more time?”

What This Project Is Not

- Not a full system implementation
- Not a penetration testing exercise
- Not a cryptography course
- Not a programming assignment

It **is** a demonstration of:

- security reasoning
- applied understanding
- professional judgement

The goal is that the report and supporting artefacts **identify, reason about, and justify security decisions in a realistic system.** - so don't forget the scenario - you are analysing and investigating a large students and staff system that has various types of user, different implementations (on-premises, cloud), different types of data.